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Review

Defense Against Adversarial Attacks: Foundations, Strategies, and Future Directions

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Abstract

Adversarial machine learning is an important area of research in computer science, focusing on understanding and mitigating attacks that make use of the vulnerabilities of machine learning models. In such attacks, adversaries aim to exploit these vulnerabilities in order to harm model's utility or violate its privacy or availability. These attacks include evasion, poisoning, exploratory, and explainability. Evasion, poisoning, and explainability attacks aim to harm the models' utility, while exploratory attacks violate the privacy of the models. To reduce the negative impacts of these attacks, a plethora of defense mechanisms have been proposed. In this survey, we review a substantial body of works and propose a comprehensive and novel taxonomy of defense strategies. We divide defense systems into eight main categories including: training-based, architecture-based, uncertainty-based, detection-based, optimization-based, transformation-based, information-theoretic, and hybrid approaches. For each category, we analyze and summarize key techniques and algorithms. Moreover, we investigate the limitations associated with these defenses, such as model generalization, robustness-accuracy trade-off, and adaptability challenges. By highlighting the strengths and weaknesses of existing defenses, this survey aims to enlighten future research towards more robust and efficient adversarial defense mechanisms.

Keywords: adversarial machine learning; adversarial defenses; adversarial attacks; defense model; detection; mitigation; robustness; privacy; utility; explainability

1. Introduction

Machine learning (ML) has revolutionized a huge number of fields by enabling systems to learn from data without explicit programming [1,2]. In healthcare, ML supports in decision making, disease diagnosis, disease risk prediction, personalized drugs, and so on [1]. Autonomous vehicles utilize ML for real-time decision-making and navigation [2]. Moreover, ML enhances natural language processing applications, such as language translation and smart assistants [3], and plays an important role in cybersecurity by detecting anomalies and preventing cyber threats [4,5]. With its recent progress, ML continues to move industries to the next level, however, this advancement has led to competitions between countries and corporations, making learning systems more vulnerable to adversarial attacks, which degrade the model's performance or harm its privacy or explainability.

Adversarial machine learning (AML) is the study of security vulnerabilities in machine learning models caused by the increased attack surface, adversarial attacks, and techniques to defend against adversarial threats. Most of the machine learning applications operate without considering the existence of malicious activities and they currently do not include any built-in defense system [4–6].

The growing attack surface has concerned governments worldwide. For instance, The U.S. Department of Homeland Security (DHS) has identified adversarial AI threats, especially evasion attacks and generative AI, as critical threats to national security [7]. Additionally, The U.S. Congress, through its Bipartisan AI Task Force, warns that adversarial AI can be used to violate civil privacy, security and rights, necessitating strong defense strategies [8].

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With the increased attack surface and its consequences a plethora of defenses have been proposed to counter adversarial attacks. While some defenses focus on detecting adversarial perturbations, others aim to mitigate their negative effects. These strategies achieve robustness through various approaches, such as modifying training data, altering model architecture, or adjusting the model's parameters. However, achieving robustness often comes at a cost, including reduced clean accuracy, increased training and inference times, and other trade-offs. Moreover, no defense strategy can guarantee a 100% robustness, as each approach has its own vulnerabilities.

Defenses against adversarial attacks can be categorized into eight main categories: training-based, architecture-based, uncertainty-based, detection-based, optimization-based, transformation-based, information-theoretic, and hybrid approaches. Training-based defenses enhance robustness by modifying the training data or the learning process. Architecture-based defenses achieve robustness by manipulating the model's architecture. Uncertainty-based defenses aim to obfuscate adversarial effects using either static or dynamic approaches. detection-based focus on identifying adversarial inputs in different stages to ignore or reject them. Optimization-based defenses provide mathematical guarantees of robustness against specific types of attacks or specific perturbation ranges. Transformation-based defenses attempt to purify inputs by reconstructing them to remove adversarial perturbations. Information-theoretic defenses use techniques like differential privacy and mutual information to prevent data leakage. Finally, hybrid defenses combine multiple strategies to achieve better overall performance.

Our contributions are as follows:

(i) We review a substantial body of work, including state-of-the-art works, and propose a comprehensive defense taxonomy. Additionally, Table 3 provides a detailed comparison of the most significant defense strategies.

(ii) We analyze the vulnerability types of the existing defense systems. Table 4 indicates the corresponding vulnerabilities for each of the significant defense systems.

(iii) We study the most important defense strategies for different learning frameworks including federated learning (FL) and transfer learning (TL) and provide future directions for efficient countermeasures.

(iv) We introduce the defense model and its key properties, offering insights into defense systems from multiple perspectives.

The remaining parts of the survey are organized as follows: In Section 2, we analyze threat model's specifications. In Section 3, we briefly discuss the main types of adversarial attacks. In Section, 4 we introduce the defense model. In Section 5, we provide a novel and comprehensive taxonomy of defense strategies and summarize significant defenses. In Section 6, we categorize the vulnerabilities of the existing defense systems. In Section, 7, we discuss challenges and future directions. Finally, in Section 8, we conclude our work.

2. Adversarial Threat Model

In this section, we briefly introduce key components of the adversarial threat model. These components describe how attacks are characterized based on the attacker's knowledge, the type of security violation, and the attack specificity [4,9,10].

Attacker's Knowledge: Adversarial attacks can be categorized based on the attacker's knowledge of the target model. In white-box settings, the attacker has full access to the model, including its architecture, parameters, and training process. In black-box settings, the attacker has no internal knowledge and can only query the model to observe outputs. Gray-box settings fall between these two extremes, in which the attacker has partial knowledge of the model or the defense mechanisms [4,9,10].

Security Violations: Adversarial attacks can lead to three types of security violations: integrity, availability, and confidentiality [4]. Integrity violations occur when the model produces incorrect outputs while remaining functional. Availability violations occur when the system becomes unreliable or unusable due to degraded performance or denial-of-service attacks. Confidentiality violations occur

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when the attacker extracts sensitive information about the model or its training data, such as through inference or extraction attacks.

Attack Specificity: Attacks can also be categorized based on their specificity. In untargeted attacks, the adversary aims to cause incorrect predictions. In targeted attacks, the adversary forces the model to produce a specific incorrect output.

3. Attack Types

Adversarial attacks can be broadly categorized into four types: evasion, poisoning, exploratory, and explainability-harmful attacks [4]. This survey primarily focuses on defense strategies and does not aim to discuss adversarial attacks in detail. For a comprehensive overview of adversarial attacks, we refer readers to our prior survey [4] and other foundational works [11–14].

Evasion Attacks: Evasion attacks occur during the inference phase. The adversary perturbs input samples to cause misclassification without modifying the training process. These attacks often rely on gradients or decision boundaries to craft adversarial examples [4,15,16]. Some classical methods include Fast Gradient Sign Method (FGSM) [17], Basic Iterative Method (BIM) [6], DeepFool [18], Carlini and Wagner (C&W) [19], Zeroth Order Optimization (ZOO) [20], Universal Adversarial Perturbations [21], Projected Gradient Descent (PGD) [22], Boundary Attack [23], Square Attack [24], and AutoAttack [25]. Evasion attacks are widely studied due to their practicality in real-world settings.

Poisoning Attacks: Poisoning attacks occur during the training phase. The adversary manipulates the training data to influence the learned model. This can be done by injecting malicious samples or modifying labels. Early work demonstrated poisoning against SVMs [26]. Later studies explored label flipping [27,28] and backdoor attacks. These attacks are critical when training data are collected from untrusted sources [4].

Exploratory Attacks: Exploratory attacks aim to extract information about the model or its training data. They do not modify the model. Instead, they rely on query access. Membership inference attacks determine whether a sample was used in training. Model inversion attacks reconstruct sensitive features. Model extraction attacks replicate the functionality of a target model. These attacks violate data confidentiality [4,29–31].

Explainability-harmed Attacks: Explainability-harmed attacks target interpretation methods such as saliency maps, SHAP values, LIME, and feature attribution techniques. The adversary tries to produce misleading explanations while keeping predictions unchanged [32–35]. These attacks are particularly dangerous in high-stakes applications, where explanations are used to make decisions.

Figure 1 illustrates the bi-level distribution of adversarial attacks including four major categories and major attacks within each category. The percentage of each group of attacks derived from more than 350 scholarly research works on adversarial attacks along with insights from our prior survey [4]. Although the current proportion of studies in exploratory and explainability attacks remains smaller compared to other categories, they constitute rapidly emerging research directions that have attracted growing attention in recent years.

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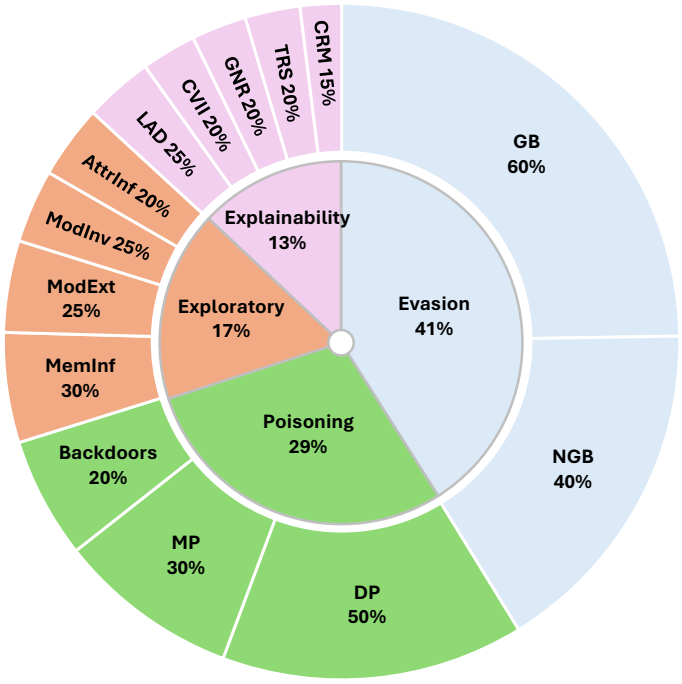


Figure 1. Bi-level distribution of adversarial attacks according to our survey of adversarial attacks [4]. The inner layer shows the proportion of major categories of adversarial attacks across the literature. The outer layer presents the relative distribution of main adversarial attacks within each category including gradient-based (GB), non-gradient-based (NGB), data poisoning (DP), model poisoning (MP), membership inference (MemInf), model extraction (ModExt), model inversion (ModInv), attribute inference (AttrInf), local attribution deception (LAD), cross-view instability induction (CVII), global narrative rewriting (GNR), triggered rationale subversion (TRS), and counterfactual and recourse misguidance (CRM).

4. Defense Model

Defense model illustrates the objective, knowledge, deployment stage, position, activation, assurance, and the protection target of a defender, highlighting different perspectives of a defense system. Figure 2 shows an overview of the defense model.

Objective: Based on the defense's objective, a defender can detect and/or mitigate the incoming threats. In detection-based objective, the only goal of the defender is to distinguish between benign and adversarial and hand over the rest of the work to a human expert, or simply reject that adversarial input. Mitigation-based defenses, on the other hand, aim to increase the resilience of the target model during different stages, so the target model can correctly predict the value of each input.

Knowledge: The defender might have complete information about the adversary's specification and is trained based on assumptions, which is unrealistic in a real-world scenario. However, having partial or no information on the attack's existence and its occurrence time can more efficiently bridge the gap between theory and practice.

Deployment Stage: A defense system can be deployed at any stage of AML lifecycle, which includes pre-training, training, post-training, testing, and deployment stages.

Position: A defense system could be either embedded within the target model or serve as an independent entity residing outside the architecture of the target model.

Activation: In a proactive-based activation, the defense module is always present in both adversarial and non-adversarial environments, while in reactive-based defense systems, the defense module starts functioning when the system is exposed to adversarial threats.

Assurance: Assurance for any defense system could be empirical or theoretical. In empirical assurance, the result is guaranteed by experimental demonstrations, while in theoretical-based assurance mathematical proofs and certifications are provided to support the evidence.

Protection Target: A defense system could protect one of the following entities: training data, algorithm, architecture, and inference/test data. In training data protection, mainly the detection-

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based strategy is useful, for algorithm and architecture protection, we have many categories such as adversarial training and neural architecture search. And the last type, usually is associated with transformation and purification techniques.

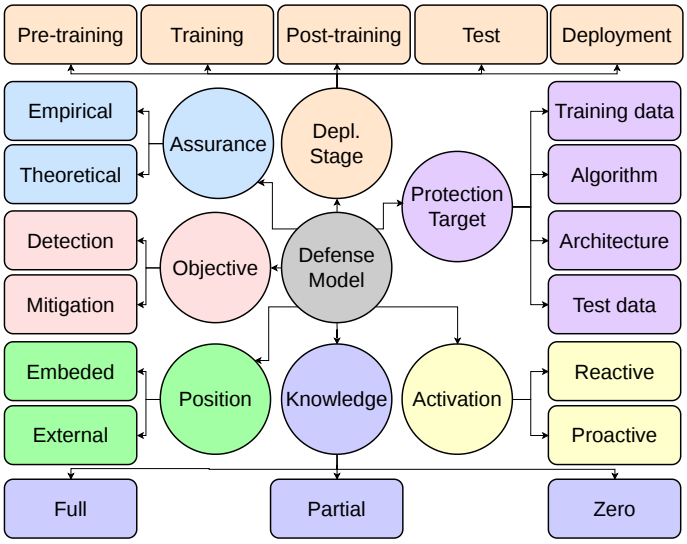


Figure 2. Defense Model.

5. Defense Strategies

We categorize the defenses into eight categories. In the following sections. Figure 3 shows a general taxonomy of the existing defense systems:

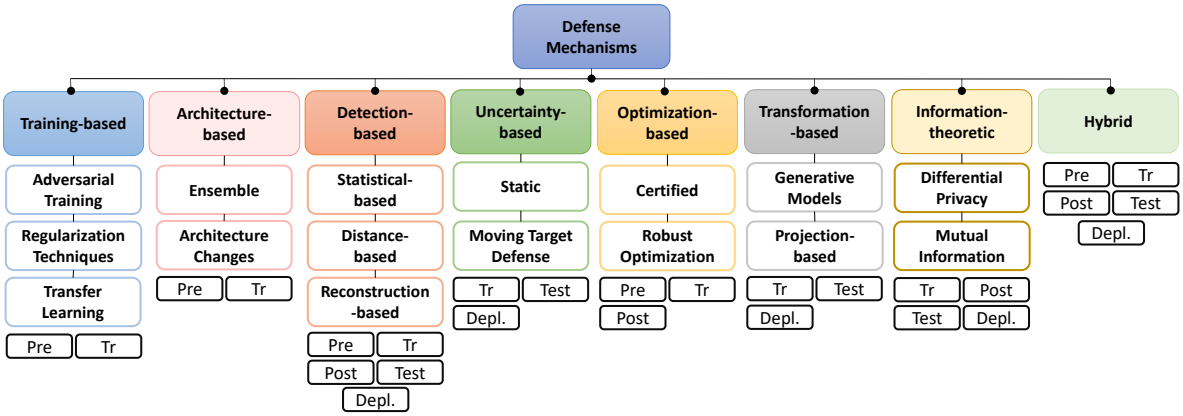


Figure 3. Taxonomy of defense mechanisms along with their placement within ML lifecycle stages - pre-training (Pre), post-training (Post), training (Tr), and deployment (Depl.).

5.1. Training-based Defenses

Training-based defenses make the model robust during training time and are divided into three categories: adversarial training, regularization techniques, and transfer learning techniques.

5.1.1. Adversarial Training

Adversarial training is a defense technique designed to improve a model's robustness against adversarial attacks. During the training session, adversarial samples are created by slightly altering the input data to maximize the model's prediction error. The model is trained using both clean (normal) and adversarially perturbed samples and learns to correctly classify both. However, this type of defense introduces some challenges including the robustness-accuracy trade-off [36–38]. There exist different types of adversarial training based on the way that adversarial samples are integrated with clean ones, and how they are generated.

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Standard Adversarial Training: In standard adversarial training, the generated adversarial samples are only from a single type of attack. Hardening Denial of Service (HDoS) [39] uses this approach in an IoT environment.

In [17], the following adversarial objective function is used during adversarial training:

$$\begin{aligned}\tilde{L}(\theta, x, y) &= \kappa L(\theta, x, y) \\ &+ (1-\kappa) L(\theta, x + \epsilon \cdot \text{sign}(\nabla_x L(\theta, x, y))),\end{aligned} \tag{1}$$

where x is a benign sample, y is the corresponding label of that sample, ϵ is the perturbation size, $L(\theta, x, y)$ is the loss function, $\nabla_x L$ is the gradient of loss function w.r.t. x, θ is the model's parameters, κ is a controlling parameter, which balances standard training and adversarial training, and $\text{sign}(.)$ returns the sign of the input. The results indicate a sharp drop in error rate against samples based on FGSM. This finding shows that an adversarially trained model is much more robust than a model without the adversarial training.

Fast adversarial training is a variant of standard adversarial training that eliminates the overhead cost of generating samples using efficient methods while retaining the efficiency by a reasonable amount [40]. Fast Free Adversarial Training (FFAT) [40] trains on the same mini batch multiple times before moving to the next one. This multiple reusing of mini batches allows adversarial perturbations to be generated iteratively without extra backpropagation. As a result, this process eliminates the computational overhead of separate gradient computations for generating adversarial samples in PGD-based adversarial training.

Trade-off Adversarial Training: These methods aim to balance robustness and accuracy [36]. TRadeoff-inspired Adversarial DEfense via Surrogate-loss minimization (TRADES) [36] is a framework that maintains the trade-off by optimizing a regularized surrogate loss:

$$\min_f \mathbb{E}\left[L(f(x), y) + \max_{x_{adv} \in B(x,\epsilon)} \frac{1}{\lambda} L(f(x), f(x_{adv}))\right], \tag{2}$$

where x_{adv} is an the adversarial sample, λ is the trade-off parameter that balances accuracy and robustness, $B(x, \epsilon)$ is a set of adversarial samples, and $\max_{x_{adv} \in B(x,\epsilon)} \frac{1}{\lambda} L(f(x), f(x_{adv}))$ improves the robustness by pushing the decision boundary away from the data.

Fair Robust Learning (FRL) [41], unlike TRADES, maintains the trade-off for each class individually, ensuring class-wise fairness rather than global balancing robustness and accuracy across the whole dataset. Helper-based Adversarial Training (HAT) [42] achieves higher accuracy and robustness in comparison to its peers.

Cascade Adversarial Training: These methods gradually improve the model's robustness by iterative injection of generated adversarial samples crafted from adversarially trained networks during their training session [43].

During adversarial training in Cascade Adversarial Machine learning Regularized with a Unified Embedding (RUE-CAM) [43], adversarial samples are generated from both the model in the training process and an already defended network. This process transfers robustness from the defended model, improving resistance to iterative attacks.

Curriculum Adversarial Training: In curriculum adversarial training, the goal is to build a curriculum of adversarial samples with different ranges of strengths. The strength gradually increases to make the model robust to stronger attacks [44].

Curriculum Adversarial Training (CAT) [44] uses batch mixing to prevent the model from forgetting weaker attacks, while making it robust against more powerful attacks. This defense uses quantization to restrict pixel values, leading to the mitigation of attack generalization. Friendly Adversarial Training (FrAT) [45], unlike CAT, can automatically increase the adversarial strength at the right time rather than using a fixed schedule.

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Guided Adversarial Training: This approach enhances the learning process of a robust model by using auxiliary guidance, often from a naturally trained model to mitigate the degradation of accuracy on clean data, while improving robustness against adversaries [46,47]. Targeted Adversarial Training (TAT) [46] navigates the robust model towards the areas with high error rates, while Learnable Boundary Guided Adversarial Training (LBGAT) [47] guides adversarial training by using logits from a clean model to shape the decision boundary of the robust model.

Multi-task Adversarial Training: The goal in multi-task adversarial training is to handle multiple objectives by simultaneously optimizing different loss functions or tasks, enabling the model to defend against a broader range of adversarial threats [48]. Hybrid Adversarial Training (HAT) [48] enhances robustness by crafting adversarial perturbations using multiple loss functions, including cross entropy (CE), feature scattering (FS), and margin (M) losses to enhance the model's robustness. The following loss function is maximized to create strong adversarial perturbations:

$$L_{\text{adv}} = \beta \cdot L_{\text{CE}}(x_{\text{adv}}, y) + \gamma \cdot L_{\text{FS}}(x, x_{\text{adv}}) + \zeta \cdot L_{\text{M}}(x_{\text{adv}}, y), \tag{3}$$

where β, γ, and ζ are tunable hyper-parameters and x_{adv} is an adversarial sample. Collaborative Multi-task Training (CMT) [49] also falls under the same umbrella.

Dynamic Adversarial Training: Here, adversarial perturbations update during the training process, and both defense and attack evolve over the time.

In Adversarially Train against Data Poisoning (ATDP) [50], the attacker and the defender are engaged in a mini-max game. The attacker poisons the data to cause mis-classification, while the defender trains a model to correctly classify poisoned and benign samples. Learn to Learn (L2L) [51], Dynamic Training Strategy (DTS) [52], and Customized Adversarial Training-based on Instance Loss (CATIL) [53] fall under the same umbrella as ATDP.

Ensemble Adversarial Training: This approach utilizes the concepts of ensemble learning in different ways. Ensemble Adversarial Training (EAT) [54] aggregates knowledge from different models by combining their adversarial outputs, while Self Ensemble Adversarial Training (SEAT) [55] combines knowledge from different training states of the same model to enhance performance.

Universal Adversarial Training: Universal Adversarial Perturbations (UAP) are independent of the inputs and can be added to any input from different datasets to cause misclassification [56,57]. To mitigate the effects of UAPs, some defenses have been proposed.

Meta Adversarial Training (MAT) [57] shares information via meta-learning approaches about optimal patches while training the model so the patches are more substantial, leading to a more robust model. MAT also encourages diversity by assigning a random target class to each meta-patch, so these meta-patches are less likely to have the same model vulnerability in common. Universal Adversarial Training (UAT) [56] and Shared Adversarial Training (SAT) [58] fall under the same umbrella as MAT; however, they do not use meta-learning concepts.

Mixed Adversarial Training: In this approach, training occurs on virtual data, which is the linear combination of samples, including both benign and adversarial, and their corresponding labels. This approach generates new training points that lie between the clean and adversarial distribution aiming to reduce the generalization error of the classifier and make the network less sensitive to corrupted labels [59,60].

Mixup [60] creates new training samples using the following linear combination:

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j, \tag{4}$$

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j, \tag{5}$$

where x_i and x_j are random samples from the original dataset, y_i, and y_j are the corresponding one-hot label encodings and $\lambda \in [0, 1]$. Then, this formula generates several virtual samples using that linear combination, and the model is trained on the virtual dataset. Regularization Mixup Adversarial

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Training (RMAT) [61] uses mixed adversarial training to protect against both membership inference and model availability attacks.

5.1.2. Regularization Techniques

These approaches regularize parameters, layers outputs, and/or reduce the complexity of the model to make it robust.

Weight Regularization: Here, the weights of the model are perturbed to make the model robust against adversarial attacks [62].

Adversarial Weight Perturbation (AWP) [62] enhances robustness by applying a double-perturbation mechanism. It first creates adversarial samples via PGD attack. Weight perturbation is calculated by identifying the worst-case direction that maximizes loss. Then, the parameters of the perturbed model are iteratively updated using Stochastic Gradient Descent (SGD) to obtain a flatter weight loss landscape and reduce the robust generalization gap.

Activation Regularization: Here, noise is added to the output of some hidden layers [63,64]. Adversarial Noise Layers (ANL) [63] adds noise into intermediate layer activations (outputs) by applying a two-round training strategy. The first forward propagation calculates loss and standard derivation of intermediate activations, while the first backward propagation calculates gradients, which are used to generate adversarial noise for activation of each layer. On the other hand, the second forward propagation computes loss based on perturbed activations, while the second backward propagation calculates gradients to update the model parameters. Finally, after training, the noise layers are removed during inference to avoid additional computational overhead.

Attribution-based Regularization: Here, the goal is to enforce robustness by constraining the stability or consistency of feature attribution/explanation maps under input perturbations. Robust Attribution Regularization (RAR) [65], Interpretability-Aware Robust Training (IART) [66], and Smooth Surface Regularization (SSR) [67] make use of this technique to mitigate explainability-harmed attacks.

Dropout: This algorithm removes hidden and/or visible units in a neural network. When a unit is dropped out, it gets isolated, and all the connections for that unit are removed. The goal of dropout is to avoid intensive co-adaptation of units by combining sub-networks' predictions that share weights [68].

Defensive Dropout Hardening against Adversarial Attacks (DDH-AA) [68] applies dropout in both training and test time to increase randomness. This method uses a drop rate r at the training time, which is optimized during the training, and for the test phase, it uses r' as drop rate, which is derived from r, on the corresponding layer, and a sub-network is sampled for it; then activation functions in the sub-network need to be scaled-up to approximately have the same prediction as the original neural network. Dropout [69] is another framework, which aims to use this approach against Membership Inference Attacks.

5.1.3. Transfer Learning Techniques

Transfer learning is the process of transferring knowledge from one classifier to another, typically for a different but related task. This approach is especially useful when the second model has limited data or resources for training [70]. Surprisingly, using transfer learning techniques can increase the robustness of the model. These techniques include task transfer learning, domain adaptation, knowledge distillation, and few-shot learning.

Task Transfer Learning: In task transfer learning, the model is trained on one task and repurposed on another task [71]. Adversarially Robust Transfer Learning (ARTL) [71] is a framework that transforms both robustness and accuracy from CIFAR-100 to CIFAR-10. The source model is adversarially trained on CIFAR-100, the feature extractor layers of the source model are frozen, and only the classification layer is retained for the target classifier to inherit the robustness. Moreover, lifelong learning methods are applied to avoid forgetting the robustness of the source model.

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Domain Adaptation: Here, unlike task transfer learning, the target model has a different domain and data distribution. Adversarial Training with Domain Adaptation (ATDA) [72] improves the generalization capability of adversarial training using domain adaptation. Defense Against Membership Inference Attacks (DAMIA) [73] aims to thwart membership inference attacks by applying domain adaptation to create a shared representation of the source domain (obfuscating dataset) and the target domain (sensitive dataset). The workflow of DAMIA consists of three phases: obfuscating dataset preparation, removing sensitive information, and domain adaptation training. In the first phase, an obfuscation dataset is prepared for the source domain by using techniques like adding Gaussian noise or motion blur. In the second phase, the labels of the sensitive data are removed. Finally, a shared representation is learned by minimizing Maximum Mean Discrepancy (MMD) distance.

Knowledge Distillation: Here, the defenses make use of knowledge distillation to gain robustness.

Defense Distillation (DD) [74] uses a new variant of knowledge distillation; instead of transferring the knowledge from a larger network to a smaller one, the target model extracts the knowledge from a network with the same architecture as itself. However, the algorithm is only applicable to the models providing energy-based probability distribution. Defense Distillation-based Mitigation Method (DDMM) [75] uses knowledge distillation to mitigate evasion attacks in NextG networks.

Few-Shot Learning: Here, the goal is to train a classifier with a good generalization ability using only a few samples [76].

Defensive Few-Shot Learning (DFSL) [76] addresses adversarial robustness in few-shot settings by transferring defense knowledge during the training session. It mitigates the distribution gap between benign and adversarial samples through feature-wise and prediction-wise consistency criteria. Masking Differential Prompting (MDP) [77] uses few-shot learning to improve the model's robustness against backdoor attacks.

5.2. Architecture-based Defenses

These defenses work on the model's architecture and dimensionality to enhance the model's robustness against adversarial attacks. These defenses are divided into two categories: ensemble defenses and architecture changes.

5.2.1. Ensemble Defense

In this approach, a set of classifiers are constructed to classify a new data point by either weighted or unweighted averaging of their predictions. An ensemble can outperform a single classifier if the ensemble system is diverse enough, and each model performs better than a random classifier [78]. Since these models consist of multiple decision boundaries, it's harder for the adversary to fool multiple decision boundaries at once rather than a single decision boundary [5,79,80].

Distillation: This technique is used to make fuzzy labels. So, instead of using hard labels, the labels are in the form of probability vectors, belonging to every class by a probability distribution [81].

The ensemble method in the defense distillation uses multiple models to decide label distribution probabilities. Defense Distillation Hardening for Random Forest (DDH-RF) [81] has two modules called Condenser and Receiver. The first one generates the probability vectors by using the training data and hard labels for random forest regressor, while the latter receives the probabilities and generates the predictions.

Hierarchical Defenses: These approaches use an ensemble of classifiers in a hierarchical manner. Application Constraints (AppCon) [79] employs multiple detector models, each focusing on identifying a specific botnet variant. The defense model uses five different applications. AppCon considers six detectors, which use five sub-detectors for the applications, and each detector is responsible for a single-botnet-variant. In other word, each detector is designed for a specific application and limits the evasion attacks by applying constraints. Model Stacking [69] uses an ensemble of random forests, logistic regression and convolutional neural network (CNN) to mitigate Membership Inference Attacks.

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Apollon [82] protects an intrusion detection against black-box and gray-box evasion adversarial threats.

5.2.2. Architecture Changes

These defenses directly work on the architecture of a model to make it robust against adversarial attacks. These defenses are divided into three categories: defensive layers, neural architecture search, and hierarchical learning.

Defensive Layers: In defensive layers, some auxiliary layers, features, or models are added inside or outside of the network to help the network to detect/mitigate adversarial attacks [83].

In Dense Layer Analysis (DLA) [83], adversarial samples are first created in a white-box manner. During feature extraction, the activation values of fully connected layers are extracted and concentrated into a sequence for every input. Then, an alarm model (a secondary neural network) is trained to classify these activation sequences as benign or adversarial. After some iterations, several alarm models are created, which can detect adversarial samples. Finally, during the test phase, the extracted features from the test set are passed to all alarm models to do the classification. If any alarm model classifies a sample as an adversary, an alarm is triggered, and a human expert is notified for further evaluation.

Scalable Certified Defense (ScaleCert) [84] is a certified defense method utilizing superficial important neurons to prune and sparsify the neural network and returns an alert if the image is predicted as an adversary. In [85], the authors remove the last layers of a trained neural networks and replace them with a random forest to gain robustness against transfer attacks.

Neural Architecture Search-based (NAS-based) Defense: NAS aims to automatically design a neural architecture with the best possible performance. NAS begins with a set of operations and a controller to obtain a large number of neural architectures based on the search space created by the operations. The candidate architectures will be trained, and, then, ranked based on their accuracy. This process will be used as feedback for the search strategy, letting new architectures emerge [86].

Neural Architecture Search for Wide spectrum Adversarial robustness (Wsr-NAS) [87] and Adversarially Robust Architecture Rush (AdvRush) [88] make use of NAS to mitigate white-box and black-box attacks.

Hierarchical Learning: This approach uses multi-level features extraction or classification to improve robustness [89,90]. Hierarchical Feature Alignment (HFA) [89] uses hierarchical learning at feature level, while Deep Learning-based Fine-grained Hierarchical Learning for Malware Classification (DL-FHMC) [91] and Hierarchical Learning Network Traffic Attack Classification (HL-NTAC) [90] use this approach at classification level.

5.3. Detection-based Defenses

These defenses mainly focus on detecting and flagging the suspicious samples during training and/or test phases. These defenses usually remove or reject those samples that are considered suspicious. Additionally, they may send those samples to a human expert for further analysis.

5.3.1. Statistical-based Detection

This strategy uses statistical models, like statistical distributions, to detect adversarial samples. This defense type includes methods such as anomaly detection, density-based, and uncertainty-based approaches.

Anomaly Detection: These defenses detect samples that do not share the same pattern as other data samples or deviate from the rest of the distribution.

Kernel Distance Local Intrinsic Dimensionality (K-LID) [92] calculates the local intrinsic dimensionality using kernel distance for every sample. Then, the model calculates the likelihood ratio of K-LID distribution of benign and attack distribution for every sample and fits an SVM on the data to predict test data labels.

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Stateful Query Detection Defense (SQDD) [93] keeps track of the history of queries by a user and determines whether or not this chain of queries behave in an adversarial manner. Cascade Classifier based on Convolutional Filter Statistics (CC-CFS) [94] uses a cascade classifier to distinguish between benign and adversarial samples. Online Anomaly Detection (OAD) [95] and Nearest Neighbors Influence Function (NNIF) [96] are other defenses falling under the umbrella of anomaly detection.

ConsistencyChecker [97] protects against explainability-harmed attacks by comparing a model's explanations to those from multiple similar models and flags them as deceptive if they deviate significantly from the expected pattern.

Density-based Detection: These approaches use density-based techniques like Gaussian Mixture Models (GMM) to find the subsets containing adversarial samples. BIC-based Mixture Model Defense (BIC-D) [98] minimizes the Bayesian Information Criterion (BIC) by removing, modifying, or retaining the components, reducing adversarial components. Class-Aware Score Network (CASN) [99] is another density-based defense for detection.

Uncertainty-based Detection: These approaches employ uncertainty-based techniques to detect adversarial samples. Däubener et al. [100] introduced uncertainty quantification techniques, including Bayesian neural networks and Monte Carlo dropout, to detect adversarial samples in speech recognition systems. Sheikholeslami et al. [101] proposed a randomized approach for adversarial detection using minimum uncertainty metrics, including non-uniform dropout, in deep neural networks. Lightweight Bayesian Refinement (LiBRe) [102] utilizes Bayesian Neural Networks to detect adversaries.

Hybrid Detection: These defenses combine multiple statistical-based approaches to detect adversaries. Density and Bayesian Uncertainty-based Detection (DBUD) [103] combines density and uncertainty estimations; density estimation detects samples that are far from the data manifold, while uncertainty captures samples in low-confidence regions, which can not be done using density estimation.

5.3.2. Distance-based Detection

This strategy applies similarity measurements to compare adversarial samples with the non-adversary peers, using either Conversion or Modification approaches.

Conversion: In conversion, samples are converted to another form. (e.g., audio to text). Multi-Version Programming-Inspired for Detecting Adversarial Examples in Audio Recognition Systems (MVP-EARS) [104] uses Automatic Speech Recognition systems (ASR). All the ASRs, including target ASR and auxiliary ASRs, receive the input audio and convert it to text. The Similarity Calculation Component checks the similarity between the target ASR text and every auxiliary text ASRs. Finally, a binary classifier receives the scores to decide whether or not the audio is an adversary. The algorithm may fail if the malicious commands in the adversarial sample and the host transcript are very similar.

Modification: The modification technique alters a sample by changing its representation or generating an alternative feature representation. The distance between this modified sample and benign samples is then computed to evaluate its similarity.

Acoustic-Decoy Detection of Audio Adversarial Examples (ADD-AAE) [105] modifies the input using low-pass filtering, 8-bit reduction and audio silence removal then compares them with the initial audio sample to decide whether or not the input is adversarial. If an adversarial sample is affected by environmental noise, which can lead to the model's accuracy degradation. Rejection-based Detection [106] also applies modification approach.

5.3.3. Reconstruct-based Detection

This approach uses generative networks, like GANs and Autoencoders, to distinguish adversarial samples from benign ones by reconstructing samples and calculating the reconstruction error.

GAN-based Reconstruction: This approach uses generative adversarial networks (GANs) for reconstructing and detecting adversarial samples.

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Poisoning Defense Generative Adversarial Network (PDGAN) [107] uses a GAN to reconstruct the training data from user updates since the participants in the federated network do not share their training data. Then, the model uses participants' models to predict the label using majority voting. By having the labels, the defense model can predict the accuracy for each participant; if a participant's model accuracy is below a threshold, it will be considered as an attacker.

Autoencoder-based Reconstruction: This approach uses autoencoders to reconstruct and detect. An autoencoder consists of an encoder and a decoder. The encoder maps the input to a lower dimension, h, while the decoder restores the dimension to the original one [108]. Feature Autoencoder for Detecting Adversarial Examples (FADetector) [108] minimizes the following reconstruction error:

$$r = \|x - g(f(x))\|, \tag{6}$$

where x is the clean input, $f(x)$ is the encoder, and $g(f(x))$ is the decoder. FADetector also combines mean squared error (MSE), mean absolute error (MAE), Kullback-Leibler (KL) divergence and Jensen-Shannon (JSKL) divergence loss functions to enhance detection:

$$L(\theta) = \lambda_1 \cdot \text{MAE} + \lambda_2 \cdot \text{MSE} + \lambda_3 \cdot \text{KL} + \lambda_4 \cdot \text{JSKL}, \tag{7}$$

where each λ_i is a weight for each loss term and θ denotes the autoencoder parameters that needs to be optimized.

MagNet [109] uses a combination of detector networks and an autoencoder-based reformer to counter evasion attacks. The detector networks identify and reject inputs that deviate significantly from the normal data manifold, while the reformer moves benign or near-manifold inputs closer to the distribution of normal data before classifying them.

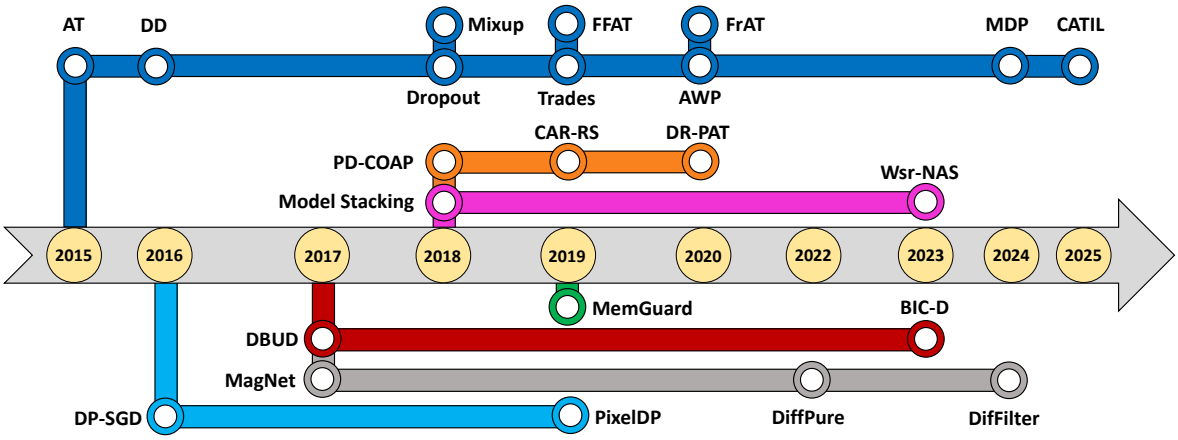


Figure 4. The timeline of the most significant adversarial defenses together with together with their major categories: blue line (training-based), orange line (optimization-based), pink line (architecture-based), cyan line (information-theoretic), gray line (transformation-based), red line (detection-based), and green line (uncertainty-based).

5.4. Uncertainty-based Defenses

This strategy tries to obfuscate and confuse the adversary by creating uncertainty. These defense types could be either static or dynamic.

5.4.1. Static Defenses

These defenses stick to fixed strategies to mislead, confuse, or obfuscate the adversaries without making dynamic or periodic changes to the system.

Bayesian Neural Networks (BNN): In Bayesian neural networks, weights w, are random variables with probabilistic distributions. By having input x and label y the posterior probability of the weights

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$p(w|x,y)$ can be calculated using $p(w)$ [110]. By using a BNN, uncertainty will propagate through the network and affect parameters and outputs, which can mislead the adversary.

Uncertainty-aware Attention Graph neural network (UAG) [111] uses a Bayesian uncertainty-aware attention technique on graph neural networks (GNN), which assigns lower weights on the attacked nodes in order to limit the data propagation between attacked nodes and other nodes (nodes with high model/data uncertainty are considered adversarial). Uncertainty-Matching Graph Neural Networks (UM-GNN) [112] uses BNN against poisoning attacks.

Deceptive: Deceptive defenses are deployed during inference/training time to mitigate exploratory attacks. These defenses use strategies such as fake sample generation, output manipulation, or noise augmentation to achieve their goal.

Generative Adversarial Network-based Defense against Model Inversion Attacks (GAN-MIA) [113] constructs two types of fake samples, inversed public samples and inversed private samples. The inverted public samples are generated from the classifier, which is trained on public data, while the inverted private samples are generated from the target victim model. To deceive the attacker, the defense maximizes the loss on inversed private samples to distort their features by minimizing the loss on inversed public samples to overfit the model on non-sensitive data, which results in the inability of the attacker that makes use of Membership Inference Attack to learn from images.

Membership Inference Guard (MemGuard) [114] adds a carefully crafted noise to every confidence score vector to obfuscate the adversary. This defense also provides formal guarantee against black box membership inference attacks.

5.4.2. Moving Target Defenses (MTD)

In Moving Target Defenses, the model becomes dynamic and tries to obfuscate adaptive and stronger attacks by dynamically changing the strategy.

Game Theory-based Defenses: Game theory provides a framework to simulate the interaction between rational players by solving a Nash Equilibrium. Stackelberg games have two players. One is the leader, and the other is the follower. The leader makes the first move, and the follower responds to that action [115].

Moving Target Defense against Adversarial Machine Learning (MTD-AML) [116] models the problem as a Stackelberg game where the defender acts as the leader. First, the defender defines π as the probability vector of deploying each algorithm. A performance metric, $transferability$, measures an algorithm's strength on attacks designed for other algorithms. The defender optimizes its strategy by solving a Stackelberg equilibrium and considering factors like switch cost and algorithms' accuracy. Edge Intelligence Moving Target Defense (EI-MTD) [117] applies a dynamic scheduling policy using Bayesian Stackelberg game.

Moving Target Defense Deep (MTDeep) [118] uses a repeated Bayesian Stackelberg game to find the optimal switching strategy between the ensemble of the classifiers. Randomized Prediction Games (RPG) [119] uses a game-theoretic framework by which the defender and the attacker select random strategies using a probability distribution.

Heuristic-based Defenses: These defenses rely on heuristics such as randomization, model diversity, retraining, and dynamic model switching rather than formal game-theoretic modeling of attacker-defender interactions. Morphence [120] is a framework that uses the most confident model in a set of diverse models. The framework refreshes the models' set after a threshold for query budget ensuring that the adversary fails. fork Moving Target Defense (fMTD) [121] dynamically generates multiple fork models by perturbing the base model and classifying by the majority of votes.

5.5. Optimization-based Defenses

These defense strategies apply optimization techniques to make the models robust against adversaries and can be divided into two categories: certified defenses or robust optimization.

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5.5.1. Certified Defenses

A classifier is certifiably robust if for any input x, a guarantee can be obtained that the prediction of the classifier is within a range around x, for instance, L_∞ or L_p ball [122]. These defenses include methods such as interval bound propagation (IBP), convex relaxation, data-dependent, and randomized smoothing.

Interval Bound Propagation (IBP): IBP propagates upper/lower bounds using interval arithmetic to train verifiably robust classifiers within a perturbation range [123,124].

Verified Robustness to Symbol Substitutions via Interval Bound Propagation (IBP-VRSS) [124] and Interval Bound Propagation for Training Verifiably Robust Models (IBP-TVRM) [123] make use of IBP to mitigate white-box evasion attacks.

Convex Relaxation Defenses: Convex relaxation is used to simplify a non-convex optimization problem by approximating it with a convex optimization problem [125]. Dual Approach to Scalable Verification (DASV) [125] formulates a verification problem as an optimization problem and solves a Lagrangian relaxation of the optimization problem to find an upper bound for the largest violation, and if the upper bound is below zero, the benignly verified property is guaranteed to hold.

Certified Defense against Adversarial Examples (CDAE) [126], Provable Defenses via Convex Outer Adversarial Polytope (PD-COAP) [127] and Certifying Incremental Quadratic Constraints via Convex Optimization (CIQC-CO) [128] are other methods using convex relaxation.

Data-dependent Defenses: This approach works on the structure of the training data in different ways and/or stages to certify robustness [129].

Certified Defenses for Data Poisoning Attacks (CDDP) [129] removes outliers and minimizes a margin-based loss on the rest of the data and computes the upper bounds for poisoning attacks by assuming that train and test distributions are close and removal does not significantly change the distribution of non-poisoned data. Local-Lipschitz Bound Training (LLBT) [130] and Certified Data Removal [131] fall under the same umbrella as CDDP.

Randomized Smoothing: Randomized smoothing is a method to construct a smoothed classifier g from a base classifier f, and by querying an input to g, the classifier returns a class which is most likely to be returned by f when the input is affected by noise. In other words, this methodology uses noise, like Gaussian noise, to smooth the model to create a robust classifier [122].

Certified Adversarial Robustness via Randomized Smoothing (CAR-RS) [122] adds Gaussian noise to the input to create smoothed classifier g, in order to make prediction's computation feasible and certify the robustness of g around the input, the method applies the Monte Carlo algorithm to satisfy both tasks. Finally, this method certifies that g is robust around the input within the L_2-radius:

$$R = \frac{\sigma}{2}\Big(\Phi^{-1}(p_A) - \Phi^{-1}(p_B)\Big), \tag{8}$$

where Φ^{-1} is the inverse Gaussian CDF, p_A is the class with the highest probability, p_B is the class with the second highest probability, and σ is a noise level controlling robustness-accuracy trade-off.

Certified Adversarial Robustness via Additive Noise (CAR-AN) [132], Certified Robustness to Label Flipping Attacks via Randomized Smoothing (CRLF-RS) [133], Practical Defense against Malware Detectors via Randomized Smoothing (PDMD-RS) [134], Chunk-based Smoothing [135], Denoised Smoothing [136], and Cost-Sensitive Randomized Smoothing (CS-RS) [137] also fall under the same umbrella. Diffusion Denoising Certified Defense (DDCD) [138] is a variant of Randomized Smoothing, which denoises the training data to provide provable robustness against clean-label poisoning attacks. Certified Robustness Control (CRC) [139] builds its core on randomized smoothing to certify robustness against adversarial inputs. Rényi Robust Smooth (RRS) [140] makes use of randomized smoothing to defend against explainability-harmed attacks.

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5.5.2. Robust Optimization

Robust optimization enhances the robustness of the model using optimization techniques, especially during the adversarial training. These techniques include min-max optimization and distributionally robust optimization.

Min-Max Optimization: Min-max optimization solves the problem by generating adversarial samples that maximize the loss function, and, then, minimizing the maximum loss to make the model robust. The formulation of the problem is in the following form:

$$\min_{f} \max_{\epsilon \in \Delta} L(f(x+\epsilon), y), \tag{9}$$

where Δ is a set of all possible perturbations.

Min Max Training for Intrusion Detection Systems (MMT-IDS) [141] and Min Max Optimization for Intrusion Detection Systems (MMO-IDS) [142] use min-max optimization for intrusion detection systems.

Distributionally Robust Optimization: This defense makes the model robust, whether by training it on different data distributions or finding an adversarial distribution for each input [143]. Distributional Robustness with Principled Adversarial Training (DR-PAT) [144] and Adversarial Distributional Training (ADT) [143] make use of this approach.

5.6. Transformation-based Defenses

This defense transforms adversarial samples into clean ones by removing adversarial perturbations and purifying the samples [145]. These defenses mainly have a denoising and reconstruction phase. In denoising, they make the input noisy, while in reconstruction, they retrieve purified images.

5.6.1. Generative Models

This approach uses generative networks like GANs, autoencoders, diffusion models, and autoregressive models to transform adversarial samples into benign ones.

Diffusion-based Transformation: Diffusion models perform purification in two phases: forward diffusion process and reverse generative process. The first phase gradually converts data to noise, while the second phase converts noise to data by denoising one step at each time [146].

Diffusion Filter (DifFilter) [147] gradually adds Gaussian noise to the adversarial sample to neutralize it in the forward diffusion. Then, in the next phase, DifFilter uses filter score model learning to gradually recover the original clean input from the noisy sample. The filter score model minimizes the following loss function:

$$L_s(\vartheta) = \mathbb{E}_{p(x_s)}\left[\|\pi(x_s, \vartheta) - \nabla_{x_s} \log p(x_s)\|^2\right], \tag{10}$$

where x_s is a noisy version of input at step s, $\pi(x_s, \vartheta)$ is the score network, ϑ is the set of parameters of the network, and $\nabla_{x_s} \log p(x_s)$ is the true gradient of log probability [147].

Dense Purification (DensePure) [146], Diffusion Purification (DiffPure) [148], Diffusion Smoothing (DiffSmooth) [149], Diffusion-based Adversarial Patch Decontamination (DiffPAD) [150], and Robust Layered Defense (ROLDEF) [151] also use diffusion-based transformations to remove perturbation from the inputs.

GAN-based Transformation: This category uses GANs to purify the inputs. Conditional GAN (cGAN) is a variant of Standard GAN, mapping an input x and random noise z and solves a min-max game using the following formula:

$$\begin{aligned} &\min_{G} \max_{D} V(D, G) = \mathbb{E}_{x,y}[\log D(x, y)] \\ &+ \mathbb{E}_{x,z}[\log(1 - D(x, G(x, z)))], \end{aligned} \tag{11}$$

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where D is the Discriminator and G is the Generator. With cGAN, it is possible to direct the data generation process to obtain the desired results [145].

Defense against Adversarial Examples using conditional Generative Adversarial Network (DAE-cGAN) [145] and Attack-Resilient GAN (AR-GAN) [152] make use of this approach.

Autoencoder-based Transformation: These defenses use an autoencoder to reconstruct input x and produce $\hat{x}$, which is the closest approximation of the original input. Then, by classifying both x and $\hat{x}$ and calculating their probability divergence, it decides whether or not the input is adversarial [153,154]. If an image is considered as adversarial, the predicted label of $\hat{x}$ can be considered as the final prediction.

Denoising autoencoder (DAE) removes adversarial perturbation from the input via image-denoising preprocessing. A denoising autoencoder performs dimensionality reduction and removes/reduces adversarial perturbations [153,154]. Sparse autoencoders apply sparsity constraints, and only some neurons are active to limit the perturbations [153].

Deep Denoising Sparse Autoencoder (DDSA) [153] is a method that combines sparsity and denoising techniques, ensuring that adversarial effects are mitigated. Gu and Rigazio [155] used DAE to remove adversarial perturbations. Robust Iterative Data Estimation (RIDE) [156] is another framework using DAE. Compression Defend (ComDefend) [157] uses a similar structure to autoencoders to purify images.

Autoregressive-based: These defenses use autoregressive models for transformation purpose. PixelDefend [158] uses PixelCNN architecture, which is a variant of autoregressive models to purify adversarial inputs.

Transformer-based: These defenses use the models built on Transformer architecture to denoise and reconstruct the inputs. In [159], the authors use BERT to reconstruct the textual inputs and empirically prove its effectiveness against a wide range of attacks. Transformer defense (Trans-defense) [160] uses cross-attention and discrete wavelet transform (DWT) to perform frequency-aware image purification.

5.6.2. Projection-based

These defenses regularize the gradients, inputs, or features in the areas with a high probability of adversarial perturbations and map them onto a data manifold [161].

Sparse Coding: Sparse coding is a projection-based approach with two components: dictionary learning and sparse coding. In dictionary learning, the most important and representative features are learned from a benign dataset, while sparse coding reconstructs the inputs by a sparse combination of dictionary filters [162]. RegionSparse [162], Sparse Transformation Layer (STL) [163], Sparse Representation Transformation Defense (SRTD) [164], and Divide Denoise Defend (D3) [165] fall under the sparse coding umbrella.

Dictionary-free: These defenses perform projection without using dictionaries. Local Gradients Smoothing (LGS) [161] computes the gradient magnitude of an input image by calculating the following distance:

$$\|\nabla x(a,b)\| = \sqrt{\left(\frac{\partial x}{\partial a}\right)^2 + \left(\frac{\partial x}{\partial b}\right)^2}, \tag{12}$$

where a and b are horizontal and vertical directions in the image plane. The gradients are then normalized using the following formula to ensure consistency across the image:

$$g(x) = \frac{\|\nabla x(a,b)\| - \|\nabla x(a,b)\|_{\min}}{\|\nabla x(a,b)\|_{\max} - \|\nabla x(a,b)\|_{\min}}. \tag{13}$$

Finally, the normalized gradient $g(x)$ is projected onto the original image to mitigate noisy perturbations in the input image domain. Dimensionality Reduction Defense (DRD) [166] uses principal component analysis (PCA) to reduce dimensionality and remove the perturbation by reconstructing the data.

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5.7. Information-theoretic Defenses

These defenses apply privacy preserving strategies such as differential privacy (DP) and mutual information (MI) strategies to reduce data leakage.

5.7.1. Differential Privacy-based Defenses

Differential privacy strategies try to mitigate information leakage by using statistical divergences. These defenses are specially effective against the attacks that violet the privacy of systems [167]. However, studies have shown that differential privacy could also be useful against evasion and poisoning attacks as well. Privacy could be preserved at multiple levels such as gradient, model, and output levels.

Gradient Level: This category adds noise to the gradients. Differential Privacy Stochastic Gradient Descent (DP-SGD) [168] was introduced in 2016 and [169] evaluated its functionality against some white-box attacks in 2021. Differential Privacy Noise Clipping (DP-NC) [170] modifies the training objective and combines differential privacy noise into the gradient computation by clipping the gradients and adding Gaussian noise to them. Gradient Shaping [171] normalizes large gradients, and adds noise to all the gradients to counter the negative impacts of the poisoned training data.

Model Level: In this category, privacy-preserving mechanisms operate across the entire model by adding noise to the layers, features, objective functions, and/or data. Pixel Differential Privacy (PixelDP) [172] defends against adversarial attacks by bounding the sensitivity of the model's predictions to p-norm changes in the input, which is achieved by adding a noise layer drawn from Laplace or Gaussian distributions to the network. During inference, PixelDP uses Monte Carlo sampling to provide certified robustness guarantees for norm-bounded perturbations.

Stochastic Batch (StoBatch) [173], Differentially Private Empirical Risk Minimization (DP-ERM) [174], and Differentially Private Recommender Systems (DP-RS) [175] are another model-level differential privacy defenses.

Output Level: This category adds noise to the output to cause obfuscation. One Parameter Defend (OPD) [176] modifies and normalizes the confidence score vectors to decrease information leakage against both membership inference and model inversion attacks. Differential Privacy Bad Data Detection (DP-BDD) [177] is an output-level DP used to detect and mitigate adversarial inputs under the Stackelberg framework.

5.7.2. Mutual Information-based Defenses

This strategy limits the information of the model input, which is contained in the output to prevent the adversary from inferring attributes and training data using the output [178]. This defense type prevents information leakage either by regularizing or optimizing mutual information.

Mutual Information Regularization (MIR): This approach uses a regularization term to minimize or maximize objective function. Mutual Information Defense (MID) [178] limits the dependency between X and Y by training the model using the following loss function and adding a regularizing term $\lambda I(X;Y)$ to the original loss function:

$$\min_{f\in\mathcal{H}} \mathbb{E}_{(x,y)\sim p_{X,Y}}[L(y,f(x))] + \lambda I(X;Y), \tag{14}$$

where X is the training data distribution, Y is the predicted label distribution, f is deterministic model, λ balances the trade-off between privacy and utility, and $I(X;Y)$ is mutual information, which equals to $H(Y) - H(Y|X)$, where $H(Y)$ is the entropy of the model output and $H(Y|X)$ is the conditional entropy of Y given X. Information Theoretic Defense (ITD) [179] and Regularization Mix (RegMix) [180] are another defense frameworks that are based on MIR.

Mutual Information Estimation (MIE): This method estimates how much information a latent representation retains about the input and maximizes/minimizes the mutual information between them.

Natural Adversarial Mutual Information-based Defense (NAMID) [181] maximizes the mutual information between input and output and minimizes adversarial mutual information during the

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training session in order to enhance the model's robustness. Mutual Information AutoEncoder Detector (MIAED) [182] uses a dual autoencoder and mutual information neural estimation to detect adversarial samples. Table 1 shows where each defense category is applicable during ML lifecycle stages. Training-based defenses include tampering with the loss function, applying regularization, or transferring knowledge, all of which are applicable during pre-training or training. Architecture-based defenses are introduced before the training session and continue through the training phase, as they involve selecting or designing robust model structures such as ensembles or neural architecture search. Detection-based defenses include removing anomalous samples before training, detecting poisoning during training, training detector models after training the base model, detecting adversarial samples during inference, or monitoring after deployment.

Moreover, uncertainty-based defenses include training Bayesian models or deploying moving target defenses during test and deployment. Optimization-based defenses tamper with the optimization procedure during training and/or apply verification after training. Transformation-based defenses include training purifier models and embedding these models with the base classifier during test or deployment to purify inputs.

Information-theoretic defenses primarily aim to preserve privacy by regularization/optimization during the training session or by adding noise during the test or deployment. Finally, hybrid defenses explicitly span multiple stages. As a result, detection-based, information-theoretic, and hybrid defenses emerge as the most practical, as they provide coverage across significant portions of the ML lifecycle.

Table 1. Placement of defense strategies across ML lifecycle stages - pre-training (Pre), training (Tr), post-training (Post), deployment (Depl.), and information-theoretic (IT).

Defense	Stage				
	Pre	Tr	Post	Test	Depl.
Training-based	✓	✓	✗	✗	✗
Architecture-based	✓	✓	✗	✗	✗
Detection-based	✓	✓	✓	✓	✓
Uncertainty-based	✗	✓	✗	✓	✓
Optimization-based	✓	✓	✓	✗	✗
Transformation-based	✗	✓	✗	✓	✓
IT	✗	✓	✓	✓	✓
Hybrid	✓	✓	✓	✓	✓

5.8. Hybrid Defenses

A hybrid defense is a combination of multiple defense techniques with the goal of better defense performance. These defenses are usually implemented in different layers of a system to provide a better protection against adversarial attacks, especially against evasion attacks.

Divide We Fall (DWF) [5] combines adversarial training with an ensemble architecture of mixture-of-experts to maintain the clean accuracy while providing protection against evasion attacks. Adversarial Defense Through Robust Bayesian Neural Networks (AdvBNN) [110] combines adversarial training with BNN using minimax formulation and solves the problem by alternating between projected gradient descent and stochastic gradient descent.

Other significant hybrid defenses include Discrete Cosine Transformation Support Vector Machine (DCT-SVM) [183], Salient Feature Extractor (SFE) [184], CardioDefense [185], and Ensemble Adversarial Training with Knowledge Distillation (EAT-KD) [186]

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Table 2. The effectiveness of each defense category against adversarial attacks. The degree of circle fill indicates effectiveness, with more filled circles representing higher effectiveness. IT in the first column stands for information-theoretic.

Defense \ Attack	Evasion	Poisoning	Exploratory	Explainability
Training-based	●	◕	●	●
Architecture-based	●	○	◑	○
Detection-based	●	●	○	◑
Uncertainty-based	◕	◔	◕	○
Optimization-based	●	●	◔	◑
Transformation-based	●	○	○	○
IT	◑	◑	●	○
Hybrid	●	○	○	○

Table 2 summarizes our analysis of the effectiveness of major defense strategies against evasion, poisoning, exploratory, and explainability attacks, based on a substantial body of work. Detection-, training- optimization-based defenses are the most effective against poisoning attacks, while uncertainty- and information-theoretic approaches provide moderate support. For exploratory attacks, training- and information-theoretic methods are the strongest, with uncertainty-, architecture-, and optimization-based approaches offering moderate effectiveness. Evasion attacks can be mitigated/detected using all defense categories. Finally, training-based defenses are the most effective against explainability attacks, with detection- and optimization-based strategies playing supporting roles.

Table 3 presents the most significant defense systems along with their specifications, key characteristics, and associated threat models. This structured comparison highlights the diversity of existing approaches and provides a unified view of adversarial defense methods.

Table 3 reveals several important observations regarding the design and evaluation of adversarial defense methods. First, the majority of existing approaches focus on the evasion attacks, indicating that test-time attacks remain the primary concern in the literature, while comparatively fewer works address poisoning and exploratory attacks. Second, most defenses are evaluated under the white-box setting, suggesting that methods are commonly designed and tested under strong adversarial assumptions, where the attacker has full knowledge of the model. Finally, the dominant application domain is computer vision, with a large proportion of defenses developed and validated on image-based tasks, whereas other domains receive relatively less attention.

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Table 3. Most significant defense systems. Column abbreviations and compact entries are as follows: **Defense category**: moving target defense (MTD), neural architecture search (NAS), interval bound propagation (IBP), mutual information regularization (MIR), mutual information estimation (MIE), bayesian neural networks (BNN), randomized smoothing (RS), and mutual information (MI). **Knowledge**: black box (B), gray box (G), and white box (W). **Target model**: random forest (RF), multi-layered perceptron (MLP), fully connected network (FCN), decision tree (DT), wide and deep (WnD), wide-resnet (WRN), support vector machine (SVM), logistic regressor (LR), linear logistic regression (LLR), neural network (NN), deep neural network (DNN), convolutional neural network (CNN), generative adversarial network (GAN), graph neural network (GNN), visual geometry group (VGG), resnet (RN), bayesian network (BN), densenet (DN), pyramidnet (PN), shufflenet (ShNet), and squeezenet (SqNet). **Domain**: computer vision (CV), natural language processing (NLP), internet of things (IoT), speech processing (SP), and cyber physical system (CPS).

Defense Model	Defense Strategy	Defense Technique	Defense Algorithm	Defender's Goal	Attacker's Type	Attacker's Knowledge			Target Model	Domain(s)
						B	G	W		
AT [17]	Training-based	Adversarial Training	Standard	Mitigation	Evasion	✗	✗	✓	CNN, Maxout	CV
FFAT [40]	Training-based	Adversarial Training	Standard	Mitigation	Evasion	✗	✗	✓	RN, WRN	CV
TRADES [36]	Training-based	Adversarial Training	Trade-off	Mitigation	Evasion	✓	✗	✓	CNN, RN, WRN	CV
FRL [41]	Training-based	Adversarial Training	Trade-off	Mitigation	Evasion	✗	✗	✓	RN, WRN	CV
RUE-CAM [43]	Training-based	Adversarial Training	Cascade	Mitigation	Evasion	✓	✗	✓	RN	CV
CAT [44]	Training-based	Adversarial Training	Curriculum	Mitigation	Evasion	✗	✗	✓	RN, DN	CV
FrAT [45]	Training-based	Adversarial Training	Curriculum	Mitigation	Evasion	✗	✗	✓	CNN, RN, WRN	CV
TAT [46]	Training-based	Adversarial Training	Guided	Mitigation	Evasion	✗	✗	✓	BERT	NLP
LBGAT [47]	Training-based	Adversarial Training	Guided	Mitigation	Evasion	✓	✗	✓	RN, WRN	CV
HAT10 [48]	Training-based	Adversarial Training	Multi-task	Mitigation	Evasion	✓	✗	✓	CNN	SP
CMT [49]	Training-based	Adversarial Training	Multi-task	Mitigation	Evasion	✓	✓	✗	CNN	CV
ATDP [50]	Training-based	Adversarial Training	Dynamic	Mitigation	Poisoning	✗	✓	✓	RN	CV
SEAT [55]	Training-based	Adversarial Training	Ensemble	Mitigation	Evasion	✗	✗	✓	RN, WRN	CV
UAT [56]	Training-based	Adversarial Training	Universal	Mitigation	Evasion	✗	✗	✓	CNN, RN	CV
SAT [58]	Training-based	Adversarial Training	Universal	Mitigation	Evasion	✗	✗	✓	WRN, FCN	CV
MAT [57]	Training-based	Adversarial Training	Universal	Mitigation	Evasion	✗	✗	✓	RN	IoT, CV
Mixup [60]	Training-based	Adversarial Training	Mixed	Mitigation	Evasion	✓	✗	✓	RN , DN, WRN, LN, VGG	CV, NLP
RMAT [61]	Training-based	Adversarial Training	Mixed	Mitigation	Evasion	✓	✗	✓	LN, RN, WRN	CV, Security
AWP [62]	Training-based	Regularization	Weight	Mitigation	Evasion	✓	✗	✓	WRN	CV
ANL [63]	Training-based	Regularization	Activation	Mitigation	Evasion	✗	✗	✓	RN, WRN, VGG, MN, PN	CV
RAR [65]	Training-based	Regularization	Attribute	Mitigation	Explainability	✗	✗	✓	NN	CV
IART [66]	Training-based	Regularization	Attribute	Mitigation	Explainability	✗	✗	✓	WRN	CV
DDH-AA [68]	Training-based	Regularization	Dropout	Mitigation	Evasion	✗	✗	✓	CNN	CV
ARTL [71]	Training-based	Transfer Learning	Task Transfer Learning	Mitigation	Evasion	✗	✗	✓	RN	CV
DAMIA [73]	Training-based	Transfer Learning	Domain Adaptation	Mitigation	Exploratory	✓	✗	✗	CNN, RN	CV
DDMM [75]	Training-based	Transfer Learning	Knowledge Distillation	Mitigation	Evasion	✗	✗	✓	CNN	IoT
DD [74]	Training-based	Transfer Learning	Knowledge Distillation	Mitigation	Evasion	✗	✓	✗	CNN	CV
DFSL [76]	Training-based	Transfer Learning	Few-shot Learning	Mitigation	Evasion	✗	✗	✓	CNN	CV
MDP [77]	Training-based	Transfer Learning	Few-shot Learning	Both	Poisoning	✓	✓	✗	RoBERTa, DART	NLP
AppCon [79]	Architecture-based	Ensemble	Hierarchical	Mitigation	Evasion	✗	✓	✗	RF, MLP, DT, WnD	Security
EMD [78]	Architecture-based	Ensemble	Hierarchical	Mitigation	Evasion	✗	✗	✓	CNN	CV
Wsr-NAS [87]	Architecture-based	Architecture Changes	NAS-based	Mitigation	Evasion	✓	✗	✓	WsrNet	CV
AdvRush [88]	Architecture-based	Architecture Changes	NAS-based	Mitigation	Evasion	✓	✗	✓	AdvRush	CV
DLA [83]	Architecture-based	Architecture Changes	Defensive Layers	Detection	Evasion	✓	✗	✓	RN, LN, kerasExM	CV, NLP, SP
ScaleCert [84]	Architecture-based	Architecture Changes	Defensive Layers	Detection	Evasion	✗	✗	✓	RN	CV
HFA [89]	Architecture-based	Architecture Changes	Hierarchical Learning	Mitigation	Evasion	✗	✗	✓	CNN, RN	CV
HL-NTAC [90]	Architecture-based	Architecture Changes	Hierarchical Learning	Mitigation	Evasion	✓	✓	✗	MLP	Security
UAG [111]	Uncertainty-based	Static	BNN	Mitigation	Evasion	✓	✗	✗	GNN	NLP
UM-GNN [112]	Uncertainty-based	Static	BNN	Mitigation	Poisoning	✓	✓	✓	GNN	NLP
GAN-MIA [113]	Uncertainty-based	Static	Deceptive	Mitigation	Exploratory	✓	✗	✓	VGG	CV
MemGuard [114]	Uncertainty-based	Static	Deceptive	Mitigation	Exploratory	✓	✗	✗	CNN	CV
MTD-AML [116]	Uncertainty-based	MTD	Game Theory	Mitigation	Evasion	✗	✓	✗	CNN, LR, SVM	Security
MTDeep [118]	Uncertainty-based	MTD	Game Theory	Mitigation	Evasion	✗	✗	✓	CNN, MLP	CV
EI-MTD [117]	Uncertainty-based	MTD	Game Theory	Mitigation	Evasion	✓	✗	✗	MN, ShNet, SqNet	CV
fMTD [121]	Uncertainty-based	MTD	Non-game Theory	Both	Evasion	✗	✗	✓	CNN	CV, CPS
Morphence [120]	Uncertainty-based	MTD	Non-game Theory	Mitigation	Evasion	✓	✗	✓	CNN	CV
K-LID [92]	Detection-based	Statistical-based	Anomaly Detection	Both	Poisoning	✗	✓	✓	SVM	CV
BIC-D [98]	Detection-based	Statistical-based	Density-based	Both	Poisoning	✗	✓	✗	SVM, LSTM, RN	CV, NLP
LiBRe [102]	Detection-based	Statistical-based	Uncertainty-based	Detection	Evasion	✓	✗	✓	RN	CV
MVP-EARS [104]	Detection-based	Distance-based	Conversion	Detection	Evasion	✓	✗	✓	SVM, KNN, RF	SP
ADD-AAE [105]	Detection-based	Distance-based	Modification	Detection	Evasion	✗	✓	✗	DeepSpeech	SP
PDGAN [107]	Detection-based	Reconstruction	GAN-based	Both	Poisoning	✗	✗	✓	GAN	CV
FADetector [108]	Detection-based	Reconstruction	Autoencoder-based	Detection	Evasion	✗	✗	✓	VGG	CV
MagNet [109]	Detection-based	Reconstruction	Autoencoder-based	Both	Evasion	✓	✓	✗	CNN	CV
IBP-TVRM [123]	Optimization-based	Certified	IBP	Mitigation	Evasion	✗	✗	✓	WRN	CV
IBP-VRSS [124]	Optimization-based	Certified	IBP	Mitigation	Evasion	✗	✗	✓	CNN	NLP
DASV [125]	Optimization-based	Certified	Convex Relaxation	Mitigation	Evasion	✗	✗	✓	DNN	CV
CDAE [126]	Optimization-based	Certified	Convex Relaxation	Mitigation	Evasion	✗	✗	✓	NN	CV
PD-COAP [127]	Optimization-based	Certified	Convex Relaxation	Mitigation	Evasion	✗	✗	✓	CNN	CV
CDDP [129]	Optimization-based	Certified	Data-dependent	Mitigation	Poisoning	✗	✗	✓	SVM	CV
CAR-RS [122]	Optimization-based	Certified	RS	Mitigation	Evasion	✗	✗	✓	RN	CV
CAR-AN [132]	Optimization-based	Certified	RS	Mitigation	Evasion	✓	✗	✓	CNN	CV
CRLF-RS [133]	Optimization-based	Certified	RS	Mitigation	Poisoning	✗	✗	✓	CNN	CV
PDMD-RS [134]	Optimization-based	Certified	RS	Both	Evasion	✓	✗	✓	MalConv	CV, Security
MMT-IDS [141]	Optimization-based	Robust Optimization	Min-Max	Mitigation	Evasion	✗	✗	✓	MLP	Security
DR-PAT [144]	Optimization-based	Robust Optimization	Distributionally	Mitigation	Evasion	✗	✗	✓	RN	CV
ADT [143]	Optimization-based	Robust Optimization	Distributionally	Mitigation	Evasion	✓	✗	✓	WRN	CV
DifFilter [147]	Transformation-based	Generative Models	Diffusion-based	Mitigation	Evasion	✓	✓	✓	RN, WRN	CV
DiffPure [148]	Transformation-based	Generative Models	Diffusion-based	Mitigation	Evasion	✓	✗	✓	RN, WRN, ViT	CV
DAE-cGAN [145]	Transformation-based	Generative Models	GAN-based	Mitigation	Evasion	✗	✗	✓	RN, DN	CV
ComDefend [157]	Transformation-based	Generative Models	Autoencoder-based	Mitigation	Evasion	✗	✗	✓	CNN, RN	CV
RegionSparse [162]	Transformation-based	Projection-based	Sparse Coding	Mitigation	Evasion	✓	✓	✗	Inception	CV
STL [163]	Transformation-based	Projection-based	Sparse Coding	Mitigation	Evasion	✓	✓	✓	VGG, RN	CV
SRTD [164]	Transformation-based	Projection-based	Sparse Coding	Mitigation	Evasion	✓	✓	✓	RN	CV
D3 [165]	Transformation-based	Projection-based	Sparse Coding	Mitigation	Evasion	✓	✓	✓	RN	CV
LGS [161]	Transformation-based	Projection-based	Dictionary-free	Mitigation	Evasion	✗	✗	✓	Inception	CV
DRD [166]	Transformation-based	Projection-based	Dictionary-free	Mitigation	Evasion	✗	✓	✗	SVM	CV
PixelDP [172]	Information-theoretic	Differential Privacy	Model Level	Mitigation	Evasion	✗	✗	✓	CNN, RN	CV
StoBatch [173]	Information-theoretic	Differential Privacy	Model Level	Mitigation	Evasion	✗	✗	✓	RN	CV
BTDR [167]	Information-theoretic	Differential Privacy	Model Level	Mitigation	Exploratory	✓	✗	✗	LLR, CNN	CV
OPD [176]	Information-theoretic	Differential Privacy	Output Level	Mitigation	Exploratory	✓	✗	✗	CNN	CV
ITD [179]	Information-theoretic	MI-based	MIR	Mitigation	Evasion	✗	✗	✓	MLP, LN, VGG	CV
NAMID [181]	Information-theoretic	MI-based	MIE	Mitigation	Evasion	✓	✗	✓	RN	CV
MIAED [182]	Information-theoretic	MI-based	MIE	Detection	Evasion	✓	✗	✓	VGG, RN, MN, DN	CV
AdvBNN [110]	Hybrid	*	*	Mitigation	Evasion	✓	✗	✓	VGG	CV
DWF [5]	Hybrid	*	*	Mitigation	Evasion	✗	✗	✓	RN	CV

6. Defense Vulnerabilities

Adaptability is the most common vulnerability among the defenses, and occurs when the performance of a defense method depends on the severity, adaptation capability, or budget of the attack.

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Additionally, this vulnerability is also valid in the scenarios, where the attacker's performance is unclear or when the adversary is unaware of the existence of the defense. Model generalization (MG) is the second most common vulnerability. It occurs when at least one of the following conditions is valid: 1) the proposed defense generates poor results on some specific target models, while it works well on some other target models, 2) the defense significantly has lower accuracy on some datasets or distributions, while working well on others, and 3) the defense can only be useful for a limited number of tasks.

Focused vulnerability occurs when the defense method mainly focuses on a set of adversarial samples or a limited number of adversarial attack types, which can cause the model to be vulnerable to other types of attack or newer ones. Moreover, the focused vulnerability is valid if the defense method focuses only on specific aspects of robustness and overlooks all possibilities or has significantly lower accuracy on some attacks with similar strength. Robustness-accuracy trade-off (RAT) occurs when the performance of the defense method degrades significantly in a non-adversary environment, while the model works well in an adversarial environment. In other words, as robustness increases, accuracy decreases.

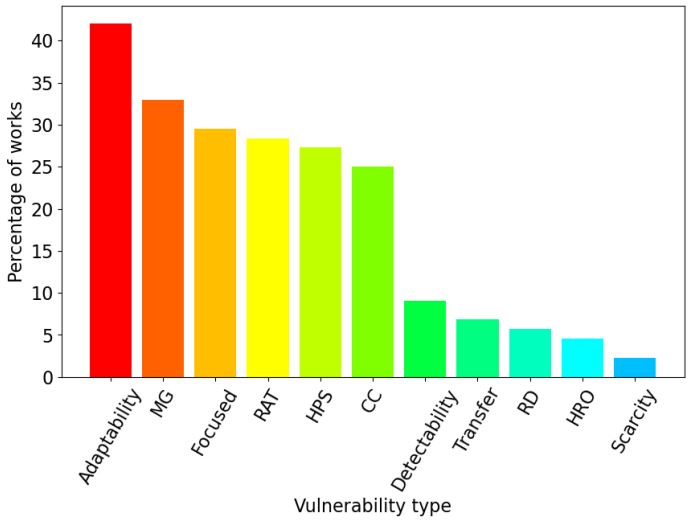


Figure 5. The proportion of each type of vulnerability observed across the research works.

Hyperparameter sensitivity (HPS) occurs when the defense method needs to carefully tune the hyperparameters to achieve high accuracy or when the impact of tuning a combination of some hyperparameters is unclear. Computational complexity (CC) occurs when the defense method requires substantial computation, which may make the defense useless in real-time settings during inference. Moreover, CC is valid when achieving better results, leads to substantially higher computational costs. Detectability occurs when at least one of the following conditions is valid: 1) the attacker can evade the defense system on some condition (e.g., high similarity to legitimate samples), 2) the distribution of benign and adversarial samples overlaps, 3) the false negative rate is significantly higher than the false positive rate, and 4) when the environment causes the detection to be impossible or more complex (e.g., environmental noises).

Transfer vulnerability occurs when adversarial samples that affect one model or algorithm can also be used against other models or algorithms. Robustness Degradation occurs when the model's robustness degrades for some attack strength ranges. High resource overhead (HRO) occurs when the defense model needs too many resources to perform well, making it infeasible to deploy on devices with low memory. Scarcity occurs when the model has few labeled samples in the training process, which is a common type of vulnerability for few-shot defense systems.

Figure 5 illustrates the percentage distribution of each vulnerability type across a substantial number of research works. As depicted, adaptability is the most frequent vulnerability type followed by model generalization. On the other hand, vulnerabilities such as scarcity occur rarely. This gap

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shows that more common vulnerabilities need to be addressed promptly in order to build more resilient defense systems.

In Table 4, we investigate the vulnerability types of the most significant defense systems. Some defense systems face several vulnerabilities, which highlights the need for hybrid and adaptive defenses to simultaneously tackle multiple vulnerabilities.

Table 4. Types of vulnerabilities of the most significant defense systems - model generalization (MG), robustness accuracy trade-off (RAT), adaptability (Adapt.), detectability (Detect.), high resource overhead (HRO), computational complexity (CC), hyper parameter sensitivity (HPS), and robustness degradation (RD).

Defense Model	Vulnerabilities										
	MG	Transfer	Adapt.	Focused	Detect.	Scarcity	RAT	HRO	CC	HPS	RD
TRADES [36]	✗	✗	✓	✗	✗	✗	✓	✗	✗	✓	✗
CAT [44]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗
FrAT [45]	✗	✗	✓	✓	✗	✗	✓	✗	✗	✗	✗
TAT [46]	✓	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗
CMT [49]	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗	✗
EAT [54]	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗	✗
SEAT [55]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗
UAT [56]	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗
MAT [57]	✗	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
Mixup [60]	✓	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗
ANL [63]	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
DD [74]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
DFSL [76]	✓	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗
MDP [77]	✗	✗	✗	✓	✓	✓	✗	✗	✗	✗	✗
AppCon [79]	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗
Wsr-NAS [87]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
AdvRush [88]	✓	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗
DLA [83]	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗
ScaleCert [84]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✓	✗
UAG [111]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗
MTD-AML [116]	✗	✗	✓	✓	✗	✓	✗	✗	✗	✗	✗
MTDeep [118]	✗	✓	✓	✗	✗	✗	✓	✗	✗	✗	✗
Morphence [120]	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
BIC-D [98]	✗	✗	✓	✗	✗	✗	✗	✗	✗	✓	✗
MVP-EARS [104]	✗	✗	✗	✗	✓	✗	✗	✗	✗	✗	✗
FADetector [108]	✓	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
IBP-VRSS [124]	✓	✗	✗	✓	✗	✗	✓	✗	✓	✗	✗
CAR-RS [122]	✓	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗
MMT-IDS [141]	✗	✗	✗	✗	✓	✗	✓	✗	✗	✗	✗
DifFilter [147]	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
DiffPure [148]	✓	✗	✗	✗	✗	✗	✓	✗	✓	✗	✗
ComDefend [157]	✗	✗	✗	✓	✗	✗	✗	✗	✗	✓	✗
MagNet [109]	✗	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
RegionSparse [162]	✗	✗	✓	✗	✗	✗	✓	✗	✗	✗	✓
STL [163]	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗
D3 [165]	✗	✗	✗	✗	✗	✗	✓	✗	✗	✓	✗
LGS [161]	✗	✗	✓	✗	✗	✗	✓	✗	✗	✗	✗
PixelDP [172]	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗	✗
BTDR [167]	✗	✗	✗	✓	✗	✗	✗	✓	✓	✗	✗
MID [178]	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗
NAMID [181]	✓	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗

7. Challenges and Future Directions

In this section we address future directions for both classical models and recent learning frameworks, where they face a larger attack surface.

7.1. Classical Models

Given the arms race nature of adversarial ML, hybrid defenses are expected to dominate. A single defense method often covers one angle of attack, but adversaries can find other vulnerabilities. By layering defenses (e.g. multiple transformations plus uncertainty checks), the system can achieve defense-in-depth, enhancing traditional defense systems. Future hybrid frameworks will be completely

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self-adaptive. Research is moving toward automated defense arrangement, where an AI agent could dynamically decide which defense techniques to use against an input or attack level. This can potentially be obtained using reinforcement learning and game theory to manage this decision process.

Hybrid approaches already show promising results, such as combining training-time and inference-time defenses to catch attacks that evade one line of defense [187]. Although hybrid defenses show some vulnerabilities, a promising future direction is to build more complex layered defense systems that integrate multiple defense strategies, mitigating each other's limitations and dynamically adapting to attacker's capabilities as the attack evolves.

7.2. Learning Frameworks

In this section, we investigate federated learning (FL) and transfer learning (TL) frameworks, focusing on the current defense strategies and potential directions.

7.2.1. Federated Learning

FL is a privacy preserving learning framework using multiple clients to train a global model. Every client has its own private data to train a local model and a server to aggregate these local models' parameter and sending back the updates to the clients. Despite advantages provided by FL, some vulnerabilities including gradient leakage provide the adversaries with more opportunities to violate privacy and security of the FL systems [4,15,188].

The threat model in FL is broader than the classical ML. This extension stems from the increased attack surface in FL, which is due to three extra entities: clients, server(s), and aggregation. The attacker could assume an insider or outsider role; an insider, unlike an outsider, can participate in the training process. Based on the location, the adversary can be on the server or the client side. Moreover, based on the adversary's behavior, it can be either malicious or honest-but-curious. A malicious adversary aims to disrupt the system's utility, while an honest-but-curious one aims to violate the system's privacy without affecting its performance [4].

Attacks: Main attack types in FL includes evasion, poisoning, data reconstruction, inference, free riders, Byzantine, and Sybil. In evasion attacks, the adversary adds some perturbations to samples misleading the deployed model [189,190]. Poisoning attacks on FL manipulate the local models during the aggregation process. These attacks are divided into two categories including data poisoning and model poisoning attacks. In data poisoning attacks, the adversary tampers with the training data. On the other hand, in model poisoning, the adversary can either manipulate the gradients/parameters from clients to the server to affect the global model, or modify the updates from the server side to the clients in order to corrupt local models [191].

Moreover, data reconstruction attacks violate the privacy of the clients. The adversary can use the statistics about the aggregation and reconstruct a part of the private data if these statistics are not differentially private [192]. In inference attacks, the adversary send queries to the global model to infer some knowledge. Inference attacks are divided into membership inference attacks (MIA), property inference attacks (PIA), and representative inference attacks (RIA) [189]. In MIA, the adversary tries to infer whether or not a sample exists in a training data [189], enabling the adversary to extract the global model by rebuilding a similar model [193]. In PIA, the goal of the adversary is to extract private patterns and/or characteristics of the training data such as data distribution. RIA aims to generate a dummy set, which is a representative for the original training data [189].

In free riders, some clients send fake local updates to the systems to access the shared global model without participating in the training process. This happens mainly due to the clients' concerns about data privacy and/or computational [188]. Byzantine attacks aim to reduce the utility of the global model by sending fake updates, which hinders the convergence of the global model [4,194]. In Sybil attacks, a malicious entity creates and controls multiple fake clients to influence the training process [4,195].

Defenses: All the defense strategies mentioned so far for classical adversarial machine learning are also valid for FL. Defenses can be launched on the server side, client side, or communication

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channel. The most important defenses in FL are adversarial training, differential privacy (DP), robust aggregation, anomaly detection, decentralized learning, and secure multi-party computation (SMPC). Adversarial training in FL is a client-side defense, where the clients train their local models on adversarial datasets to mitigate evasion and data poisoning attacks [196]. DP in FL is divided into two categories: central differential privacy (CDP) and local differential privacy (LDP). In CDP, the server is responsible for privacy preserving by adding noise to the updates, while in LDP, clients add noise to the data before sending it to the server [188]. Robust aggregation algorithms, choose the most honest clients' gradients for aggregation using different techniques [189].

Additionally, anomaly detection can be performed on both clients and data. In client anomaly detection, the system detects abnormal model weight updates and counters the negative impact caused by these clients. In data anomaly detection, the suspicious samples are removed from the training data [189]. The first method is applicable to the server side, while the latter one can be employed on client side to mitigate poisoning attacks. Moreover, there are two types in decentralized learning: blockchain-based FL and decentralized federated learning (DFL). In blockchain-based FL, clients upload their gradients to associated miners to broadcast the gradients after cross-verification. After that, the first miner propagates its block. Ultimately, all the clients download the block from their miners and update the global model. In DFL, clients update their models by communicating with their on-hop neighbors and aggregating the results from them [189].

SMPC uses cryptographic techniques to preserve the privacy of clients' data and model updates. As a result, the server can only observe aggregated results rather than individual updates. This defense helps protect against reconstruction attacks, reducing the risk of data leakage [197].

Future Directions: Despite recent progresses in securing FL systems, existing defenses are still ineffective against more complex adversaries capable of deploying multiple attack strategies. Two key future directions can enhance FL security by improving robustness and reducing the attack surface. Most existing FL defenses focus on mitigating/detecting specific attack types, such as model poisoning, or evasion attacks. However, real-world adversaries usually lunch multiple attack types at the same time, taking the advantage of the lack of complex layered defensive mechanisms. Future research should explore defense frameworks that dynamically adapt to a wide range of attack types, while preserving privacy and model efficiency.

Potential research directions could include multi-layered defense strategy, especially combining adversarial training, differential privacy, and robust aggregation mechanisms, to enhance security and privacy at multiple levels. Another promising direction is proactive security mechanisms that minimize the exposure of FL systems to adversarial environments. For instance, deploying honeypot servers and combining them with game theoretical approaches to attract adversarial clients, capturing their attack patterns without affecting the main FL model. Insights gained from these interactions can improve security and privacy.

7.2.2. Transfer Learning (TL)

In TL vulnerabilities can transfer from the source model or emerge during the transferring process. Attackers can craft adversarial samples on a base model that transfer to the fine-tuned model, which is going to be deployed in a target system [198,199].

Moreover, if the base model is from an untrusted source, it might have a Trojan horse with backdoors. Another concern is that standard adversarial training during the fine-tuning stage might not fully protect the model if blind spots exist. To address these challenges, future defenses in transfer learning will focus on robust knowledge transfer with more attention on the initialization. One idea is to perform robust pre-training, ensuring the source model is adversarially trained or certified on a wide range of adversarial inputs. Studies have shown that adversarial robustness can be improved by robustifying the pre-training and/or the fine-tuning stage [200].

A promising future direction is meta-learning for robust transfer learning, by using meta learning concepts, model's adaptation to a new task would be faster and more effective. Moreover, adversarial training with meta learning is more robust and has a higher accuracy than plain adversarial

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training [201]. By combining strong pre-training strategies, meta-learning, and robust fine-tuning procedures, future transfer learning environments can preserve security and privacy from start to finish.

8. Conclusions

Adversarial attacks remain a significant challenge to the security, privacy, integrity, and reliability of machine learning systems. Although numerous defense mechanisms have been proposed, none guarantee a complete solution due to inherent trade-offs and vulnerabilities. This survey categorized defenses into eight main categories, providing a comprehensive analysis of these techniques, and their effectiveness and limitations. The findings highlight that while training-based and architecture-based approaches can significantly enhance robustness, they often come at the cost of increased complexity and reduced accuracy in non-adversarial settings. On the other hand, detection-based and uncertainty-based methods provide promising results in identifying and/or mitigating adversarial samples but cannot adapt to different adversarial scenarios.

Some vulnerabilities such as adaptability and model generalization occur more frequently, while vulnerabilities such as scarcity are rare. This gap shows that critical vulnerabilities need to be thoroughly addressed to develop more resilient defense systems. Additionally, future research must focus on addressing these limitations by developing hybrid defense mechanisms that balance robustness, efficiency, and adaptability. By progressing the understanding of defense mechanisms, the machine learning community can implement more secure and reliable learning systems capable of resisting a wider range of adversarial attacks.

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